

CSE108: Object Oriented Programming Language Sessional

Term: July 2025

Online-5

Time: 40 Min

Delivery System

Build a Java program that models a delivery system that supports two types of delivery: Food Delivery and Parcel Delivery.

1. Base class for Delivery Request (DeliveryRequest.java)

Attributes:

- A unique request ID (`requestId`)
- The customer's name (`customerName`)
- The delivery distance in kilometers (`distanceKm`)
- A base service charge (`baseFare`)

Methods:

- A constructor to initialize attributes
- An abstract method to calculate total cost, `calculateTotalCost()`
- A concrete method to print an invoice, `printInvoice()`

Generic Delivery Rules:

1. The request ID must not be null or empty.
2. The customer name must not be null or empty.
3. The delivery distance must be greater than zero.
4. The base service charge must not be negative.

2. Derived Class for Food Delivery (FoodDelivery.java)

Additional Attributes:

- The number of food items ordered (`itemsCount`)
- A flag indicating whether the order is a late-night order (`isLateNight`)

Food Delivery Rules:

1. The number of food items must be between 1 and 20.

2. If a delivery is late at night, the delivery distance must be less than 10 km

Cost Calculation Logic (Food):

`cost = baseFare + (distanceKm × 12) + (itemsCount × 5) + lateNightFee`

`lateNightFee` is 30 if late-night, otherwise 0

3. Derived Class for Parcel Delivery (ParcelDelivery.java)

Additional Attributes:

- The weight of the parcel in kilograms (`weightKg`)
- A flag indicating whether the parcel is fragile (`isFragile`)

Parcel Delivery Rules:

1. Parcel weight must be between 0.1 kg and 30 kg
2. Fragile parcels above 10 kg are not allowed.

Cost Calculation Logic (Parcel):

`cost = baseFare + (distanceKm × 10) + (weightKg × 20) + fragileFee`

`fragileFee` is 50 if fragile, otherwise 0

4. Exception (InvalidRequestException.java)

Create one custom checked exception that is thrown (with a clear and appropriate message) whenever any generic rule or delivery-specific rule is violated.

5. Code Skeleton and Implementation Instructions

A partial code skeleton is provided. Students must complete the implementation of the provided classes according to the specifications given above.